The Geometry of Sensor Information Utilization in Nonlinear Feedback Control of Vehicle Formations

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Summary. This paper addresses the problem of distributed motion control of highly structured formations of robot vehicles. Feedback laws based on relative distance and bearing measurements are used. There are topological obstructions to globally stable system operation for such laws, but by appropriately switching among them, it is possible to stably control a rich class of formation motions.

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1 Introduction

This paper is concerned with the interplay between nonlinear control designs for the coordinated motion control of groups of robot vehicles and the use of available real-time information. It is largely motivated by recent work on coordinated motion control carried out in a laboratory setting. It is also motivated by an emerging body of research concerned with the control of highly structured motions of formations of robot vehicles. (See, for instance, [1],[8],[9],[10],[11],[18], [15],[19], and [20].) Our interest in highly structured motions has been sparked by motion control applications in which each vehicle's position must be known with a high degree of certainty at each instant of time. Beginning with [9], a number of researchers ([1],[10], and [11]) have proposed using results in graph rigidity as a framework for studying highly

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structured motions. Rigidity theory immediately suggests strategies for parsimonious use of sensor data in that it identifies minimal sets of real-time measurements needed to maintain fixed relative distances between all vehicles in the formation. Along side the goal of using available real-time data in the most parsimonious possible fashion, a great deal of current research is aimed at developing strategies for reorganizing patterns of information acquisition and utilization in response to dynamically changing operating conditions. For instance, Langbort *et al.* have proposed adapting classical $\mathcal{H}^\infty$ techniques to a class of spatially distributed systems in a way that takes account of of interconnections and makes contact with recent results on spatial invariance and control of distributed systems with nearest neighbor coupling. Eren *et al.* ([10]) have used results in rigidity theory to develop strategies for executing controlled reconfigurations of vehicle formations. Saber and Murray ([20]) have used the machinery of graph theory to address the problem of determining formation leadership. In the present paper, ideas along these lines lead to the development of a catalogue of feedback control laws which utilize a common set of relative position measurements to achieve a vairety of motion control objectives. A partial lisitng of the laws in this catalgue is given below in Section IV.

While a variety of algebraic results on patterns of information utilization have been reported, and while many of these results can be immediately translated into motion control laws for holonomic point robots (See [1].), applications of the results to the control of robot vehicles which move subject to nonholonomic constraints are much less straightforward. Indeed, while current sensing technologies provide robot vehicles with a rich capability for acquiring information about both absolute position and orientation and relative position and bearing with respect to peer vehicles, the most straightforward uses of this information in controlling motion kinematics lead us to confront the topological nonstabilizability result of Brockett (See [4], p. 195.). The aim of the present paper is to analyze a number of control laws which use relative position and bearing measurements to control nonholonomic vehicles along the lines suggested in [11]. As suggested by Brockett's theorem, none of the laws generates a globally asymptotically stable motion, but it will be shown that each law does provide desirable asymptotic performance in certain parameter ranges and in certain operating domains. A primary goal of our research is to develop formal strategies for concatenating such motions in a way that ensures that global motion objectives are achieved by control laws which produce stable motions over the time intervals and over the spatial domains in which they operate. One might think of this research as being aimed at creating a *motion description language* (MDL; see [17].) for formations in which the motion primitives are assembled into motion strings according to *topologically inspired grammatical rules*.

The recent literature cited above provides guidelines for parsimonious use of information in creating and maintaining highly structured formations of robot vehicles. The common focus is on peer sensing strategies which may

be modeled in terms of *isostatic* digraphs. Recall, that an isostatic graph is rigid and has the additional property that the removal on any edge results in a graph which is not rigid. (See [13] and [14] for basic definitions regarding graph rigidity.) The theory of graph rigidity is constructive, and Henneberg's Theorem states that a planar graph is isostatic if and only if it can be constructed from a single edge by a sequence of *vertex extensions* and *edge splits*. ([14]) Both these constructive operations have been explored in the context of formation dynamics ([10],[11]), but for the methods of creating and controlling the motions of formations discussed in the present paper, the relevant constructive procedure involves a vehicle joining a formation by positioning itself at prescribed distances from exactly two vehicles already in the formation. This is formally equivalent to the vehicle positioning itself at a prescribed distance and bearing with respect to a single vehicle already in the formation. These constructive motions realize the abstraction of vertex extension, and in terms of a general class of point robot (holonomic) models, the following result characterizes the types of rigid formations that will be of interest in the remainder of the paper:

Theorem 1. ([1]) A formation is stably rigid under distributed relative distance control if and only if the formation graph can be constructed from a single directed edge by a sequence of vertex extensions.

The paper is organized as follows. Section II states Brockett's nonstabilizability theorem and illustrates this with a standard example of a system with state space $\mathbb{R}^3$ for which no continuous asymptotically stabilizing feedback laws can be constructed. In terms of this example, we show how to define simple state feedback laws which steer any initial condition not contained in a low dimensional singular set asymptotically into any specified neighborhood of the origin. This example provides a prototype of the motion control laws studied in the remainder of the paper. In all cases, asymptotic stabilization is not possible, but by proper characterization of the operating domain, our proposed feedback can be shown to asymptotically achieve specified objectives. Section III describes in detail several feedback laws for motion control which are of use in creating the types of stably rigid formations of nonholonomic vehicles discussed above. The qualitative features of motions produced by each law will be analyzed in detail, and from this analysis, it will be possible to characterize the operating domain on which the law can be employed. Laws determining motions based on relative configuration sensing with respect to both stationary and moving coordinate frames will be described. Section IV will provide a brief discussion of how a number of laws of the type discussed in Section III can be used as primitives in a motion description language to create formations and define formation motions which are both highly structured and stable. The paper concludes with a brief description of our continuing research in Section V.

Notational convention: Throughout this paper, we shall have occasion to discuss the line-of-sight angle of one robot vehicle in terms of the body

coordinate frame of another. It will facilitate our analysis to use the notation $\arctan_2(x, y)$ to denote the unique angle $\phi \in (-\pi, \pi]$ such that $\tan\phi = y/x$.

2 Brockett's Theorem and asymptotic controllability of drift-free nonlinear systems

It is a consequence of a theorem published by R.W. Brockett in 1983 ([5]) that for certain broad classes of nonlinear control systems, it is never possible to design smooth globally asymptotically stabilizing feedback control laws. For the purposes of this paper, the theorem is as follows.

Theorem 2. *Consider a controlled nonlinear differential equation $\dot{x} = f(x, u)$ with $f(x_0, 0) = 0$, and $f(\cdot, \cdot)$ continuously differentiable in a neighborhood of $(x_0, 0)$. A necessary condition for a continuously differentiable control law $u = u(x)$ for asymptotically stabilizing x_0 is that the mapping $\gamma : N \times \mathbb{R}^m \to \mathbb{R}^n$, N a neighborhood of the origin, defined by $\gamma : (x, u) \mapsto f(x, u)$ should be onto an open set of the origin.*

An example system to which the theorem applies is the so-called Brockett integrator

$$\begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{pmatrix} = \begin{pmatrix} u \\ v \\ vx - uy \end{pmatrix}. \tag{1}$$

With control inputs $u = v = 0$, the origin $(x, y, z) = (0, 0, 0)$ is an equilibrium point, but for any $\epsilon > 0$, the point $(0, 0, \epsilon)$ has no preimage under the mapping γ. The consequence of this is that there is no smooth feedback control $u = u(x, y, z)$, $v = v(x, y, z)$ which steers trajectories with initial conditions on the z-axis to the origin. Nevertheless, the system (1) is controllable ([4], p. 179), and in fact it is known that if controls u, v are periodic functions of period $T > 0$ such that $(x(T), y(T)) = (x(0), y(0))$, then $z(T) = z(0) \pm 2A$, where A is the area of the closed curve traced by $(x(t), y(t)$ over one period, and the choice of $+$ or $-$ sign is made in accordance with whether the curve is traced in the positive (counterclockwise) or negative (clockwise) sense. Hence, motions steering trajectories between any two points on the z-axis are possible to construct. In particular, any point $(0, 0, \epsilon)$ can be steered to the origin in any finite time.

Many researchers have studied ways in which the hypotheses of Brockett's theorem can be relaxed so that various types of feedback control laws (time varying, discontinuous, etc.) can be designed to asymptotically stabilize rest points in systems like (1). (E.g. [7].)

Motivated by this observation, we consider feedback control designs for (1) which produce oscillations in the variables $x, y, \dot{x}, \dot{y}$ which decay asymptotically to zero as time increases. Specifically, consider

$$\begin{aligned} u(x,y,z) &= -k_1 x + k_2 yz, \\ v(x,y,z) &= -k_1 y - k_2 xz. \end{aligned} \tag{2}$$

Letting $r = \sqrt{x^2+y^2}$, it follows that r satisfies $\dot{r} = -k_1 r$, so that $r(t) = e^{k_1 t} r_0$. Then under the feedback law (2), z satisfies $\dot{z} = -k_2 e^{-2k_1 t} r_0^2 z$. From this it is straightforward to show that $z(t)$ tends asymptotically toward

$$z_\infty = exp[-\frac{k_2 r_0^2}{2k_1}]z_0. \tag{3}$$

Thus, while points on the z-axis cannot be steered asymptotically to zero under any smooth feedback law, by appropriate choice of the gain ratio k_2/k_1, the asymptotic limit of $z(t)$ as $t \to \infty$ can be made arbitrarily close to 0. This simple example and observation provide the basic idea which is pursued in defining motion control laws for the robot vehicles in the remainder of the paper.

3 Creating formations of vehicles subject to nonholonomic constraints

We begin this section by describing the kinds of feedback control laws which can be applied to an interesting class of wheeled robot vehicles. Because the vehicles can only move subject to nonholonomic constraints, and because only certain types of data from sensors is available, we shall be studying a very particular class of dynamical systems. Specifically, we shall treat controlled systems of the form

$$\begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{pmatrix} = \begin{pmatrix} v\ \cos\theta \\ v\ \sin\theta \\ \omega \end{pmatrix}, \tag{4}$$

where the position and orientation of a wheeled vehicle are given by its Cartesian coordinates (x, y) and orientation angle θ with respect to a fixed base frame, and the relative distance and bearing with respect to a *goal frame* are respectively ρ and φ. In equation (4), v and ω represent respectively the forward velocity and turning velocity. These are regarded as control inputs. This gives the kinematics of a knife edge, the simplest of all nonholonomically constrained vehicles.

The problem we wish to consider is that of creating a rigid formation of the type described in the Introduction. Our goal is to create such formations using using sensory feedback control laws of the form $v = v(\rho, \phi)$ and $\omega = \omega(\rho, \phi)$. The main idea is that by concatenating a sequence of such motions, formations of arbitrary complexity can be created.

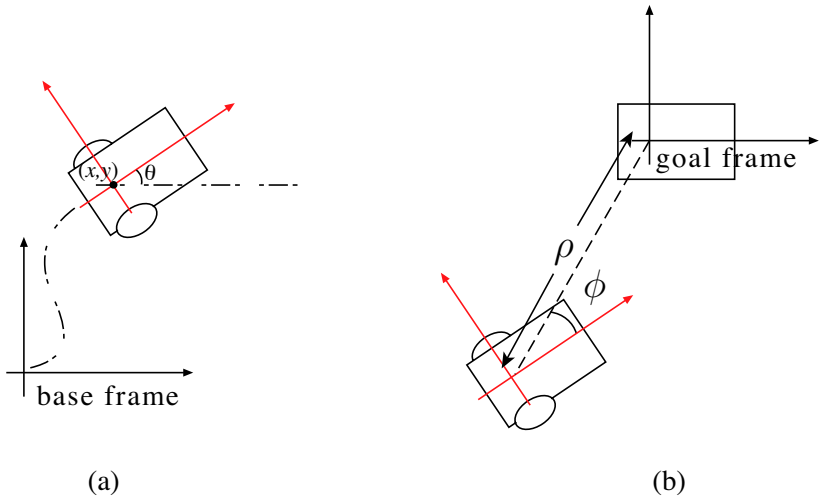

Fig. 1. (a) Kinematic variables x, y, θ give position and orientation of the vehicle with respect to some specified *base frame*. (b) The variables ρ and φ are sensor-measured position and line-of-sight direction from the vehicle to some *goal frame*.

3.1 Simple vehicle motion toward a stationary goal frame

We define our control law in terms of the variables depicted in Figure 1(b). Letting (x_r, y_r, θ_r) represent the position and orientation of the goal frame and letting ρ be the line-of-sight distance and ϕ the angle relative to the controlled vehicle's frame, it can be shown that in terms of ρ and ϕ, the equations of motion (4) are written

$$\begin{pmatrix} \dot{\rho} \\ \dot{\phi} \\ \dot{\theta} \end{pmatrix} = \begin{pmatrix} -v\,\cos\phi \\ -\omega + \frac{v}{\rho}\sin\phi \\ \omega \end{pmatrix}. \tag{5}$$

We wish to find a feedback law which will steer any initial state (x_0, y_0, θ_0) to the goal state (x_r, y_r, θ_r). Although there is no continuous feedback law which will be globally asymptotically stabilizing, we propose the following feedback law

$$\begin{aligned} v &= k_1\rho, \quad \text{and} \\ \omega &= k_2\phi + k_1 \sin\phi, \end{aligned} \tag{6}$$

with the hope of tuning the parameters k_1 and k_2 to meet our motion objectives. Using this v and ω in equation (5), we obtain closed loop equations in which the equation for ϕ becomes:

$$\dot{\phi} = -k_2\phi,$$

so that

$$\phi(t) = \phi_0 e^{-k_2 t}, \tag{7}$$

where $\phi(0) = \phi_0$. The equation for ρ becomes

$$\dot{\rho} = -k_1 \rho \cos\phi. \tag{8}$$

Since $\phi(t)$ tends exponentially to 0 as $t \to \infty$, ρ also tends asymptotically to 0. Thus, for any choices of $k_1, k_2 > 0$, the pair (ρ, ϕ) is asymptotically stable with $(\rho, \phi) \to (0, 0)$ as $t \to \infty$. The θ component of the equations requires a more detailed analysis. Under this feedback law,

$$\dot{\theta} = k_2 \phi + k_1 \sin\phi,$$

from which it follows that

$$\theta(t) = \theta_0 - \left[\phi_0 e^{-k_2 t}\right]_0^t - \left[\frac{k_1}{k_2} Si(\phi_0 e^{-k_2 t})\right]_0^t,$$

where $Si(x)$ is the Sine Integral defined by

$$Si(x) = \int_0^x \frac{\sin t}{t}\, dt.$$

We wish to have $\theta(t) \to \theta_r$ as $t \to \infty$, so that

$$\theta_r = \theta_0 + \phi_0 + \frac{k_1}{k_2} Si(\phi_0). \tag{9}$$

From equation (9), we see that the gains need to be chosen to satisfy the ratio

$$\frac{\theta_r - \theta_0 - \phi_0}{Si(\phi_0)} = \frac{k_1}{k_2}. \tag{10}$$

It is also necessary that both k_1,$k_2 > 0$. There are three cases to consider:

Case $-\pi < \phi_0 < 0$. In this case, since Si is an odd function, we must choose the representation of our goal orientation θ_r such that $\theta_r - \theta_0 < \phi_0 < 0$. This will ensure that both $k_1, k_2 > 0$, and the resulting trajectory will have the vehicle turning in the appropriate (clockwise) direction.
Case $0 < \phi_0 < \pi$. In this case, we must chose the representation of our final goal orientation such that $\theta_r - \theta_0 > \phi_0$, and the resulting motion has the vehicle turning monotonically in the counter clockwise direction.
Case $\phi_0 = 0, \pi$. In this case, the vehicle approaches (moves away from) $(x_r, y_r) = (0, 0)$ along the line-of-sight vector $(x_r - x_0, y_r - y_0)$ $(\phi \equiv 0)$, and we see that the control law (6) cannot be used to steer the vehicle into the desired configuration.

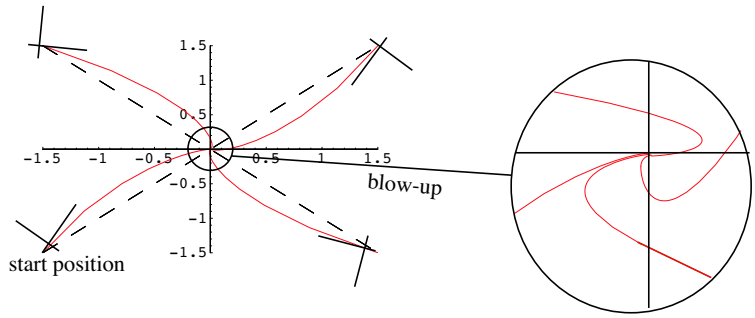

Fig. 2. The controlled vehicle starting at four different points on forty-five degree lines and moving to a goal configuration with coordinates $(x, y, \theta) = (0, 0, 0)$. Although it is difficult to see even in the magnified diagram, the vehicle starting in the upper left-hand quadrant must turn through more than $7\pi/4$ radians.

The content of this analysis is illustrated in Figure 2. We plot four motions, starting at four different points which are equidistant from the origin along the diagonal lines indicated. In all cases, the initial heading relative to the line-of-sight to the goal is $\phi_0 = -\pi/10$, so that the vehicles turn in a clockwise direction as they move toward the goal. Since in all cases, the goal configuration has the vehicle's body axes aligned with the given spatially fixed frame, the different initial configurations require different amounts of turning—with the vehicle starting in the lower left-hand quadrant turning the least and the vehicle in the upper left-hand quadrant turning the most. The gains have been chosen so that in all cases $k_1 = 1$ and $k_2 = 0.397813,\ 0.132604,\ 0.0795626$, and 0.0568305 respectively corresponding to the initial conditions in the lower left-hand quadrant and proceeding counter clockwise to the upper left-hand quadrant. Explicit calculation shows that in all cases $(\rho(t), \phi(t), \theta(t)) \to (0, 0, 0)$ as $t \to \infty$, and that $\dot{\rho}/\dot{\phi} \to 0$, so that the curvature of the paths increases as $t \to \infty$. Moreover, the relative rates of increasing curvature are such that the curves never intersect.

Remark 1. On the gain parameters. The path followed by (5) under the control law (6) depends only on the ratio k_1/k_2, while the speed with which the vehicle traverses the path depends on the actual magnitudes of k_1 and k_2.

Remark 2. An example of hybrid feedback control. For any choices of $k_1, k_2 > 0$, ρ and ϕ tend asymptotically to zero. If (10) is satisfied exactly, θ also tends asymptotically to zero. The real-time data needed to realize this control comes from sensor measurements of ρ and ϕ. Our control of θ is accomplished by choosing k_1 and k_2 to satisfy (10), and in principle, this needs to be done only once—before initiating the motion.

Remark 3. An invariant quantity. Suppose $(x(t), y(t), \theta(t))$ is a trajectory defined by the control law (6). Suppose the motion is stopped at some time

T, $0 < T < \infty$. We then restart the motion, using the same strategy with

$$\frac{\theta_r - \theta(T) - \phi(T)}{Si(\phi(T))} = \frac{k_1}{k_2}.$$

This ratio must be exactly what it was on the first segment of the trajectory, and this shows that in fact

$$\frac{\theta_r - \theta(t) - \phi(t)}{Si(\phi(t))}$$

is invariant under the motion of (5) with respect to the control (6). This in turn, allows us to write down an equivalent feedback law

$$v = \rho, \quad \text{and}$$

$$\omega = \frac{Si(\phi)\ \phi}{\theta_r - \theta - \phi} + \sin\phi,$$

which, for the proper choice of θ_r representing the goal configuration, will steer (5) asymptotically to $(\rho, \phi, \theta) = (0, 0, \theta_r)$, provided the initial condition $\phi_0 \neq 0$.

Remark 4. Sensitivity and asymptotic behavior. Since the initial relative orientation $\theta_r - \theta_0$ is assumed to be difficult to measure accurately, it is of interest to determine how sensitively the value of $\theta(t)$ depends on k_1, k_2 in the asymptotic limit as $t \to \infty$. From equations (9) and (10), we find the magnitude of $\frac{\partial \theta_r}{\partial k_2}$ is

$$\frac{(\theta_r - \theta_0 - \phi_0)^2}{|Si(\phi_0)|}.$$

When the difference between the initial and goal orientation $|\theta_r - \theta_0|$ is large and $|\phi_0|$ is small, the sensitivity can be quite large. It is also interesting to note that when $|\theta_r - \theta_0|$ is large and $|\phi_0|$ is small, the asymptotic curvature of the trajectory defined by the feedback law (6),(10) is <u>infinite</u>. The conditions for this phenomenon can be explicitly stated and are easily proved using closed-form expression for solutions to the closed-loop equation (5) with v and ω given by (6). Specifically, we write the solution of the ρ-component of the (5) as

$$\begin{aligned}\rho(t) &= \exp[-k_1 \textstyle\int_0^t \cos\phi(s)\, ds] \\ &= \rho_0 \exp[\tfrac{k_1}{k_2} \textstyle\int_{\phi_0}^{\phi(t)} \tfrac{\cos u}{u}\, du]\end{aligned}$$

where $\phi(t)$ is given by (7). Thus, using the controls (6), the curvature of the trajectory generated by (7) at time t is

$$\frac{\omega(t)}{v(t)} = \frac{k_2 \phi_0 e^{-k_2 t} + k_1 \sin[\phi_0 e^{-k_2 t}]}{k_1 \rho(t)}.$$

Invoking l'Hôpital's rule, it can be shown that

$$\lim_{t\to\infty}\frac{\omega(t)}{v(t)} = \begin{cases} \infty & \text{if } k_1 > k_2, \\ \frac{(k_1k_2+k_2^2)\phi_0}{k_1^2\rho_0} & \text{if } k_1 = k_2, \\ 0 & \text{if } k_2 > k_1. \end{cases}$$

Viewed in terms of the definitions of k_1, k_2 in (10), this result indicates that for a large open set of problem boundary conditions $\theta_r - \theta_0$, ϕ_0, the asymptotic curvature using (6) is infinite. Indeed, for all trajectories depicted in Figure 2, the asymptotic curvature is infinite. There is no simple characterization of the detailed geometry of these paths in the limit as $t \to \infty$, and because large excursion away from the goal frame are possible, paths in which $k_1 > k_2$ are deemed to be undesirable. (See e.g. the path in Figure 2 which originates in the upper left hand quadrant.)

Figure 3 provides a caricature of paths which have asymptotically finite curvature in their approach to the goal frame. Such paths may be characterized in terms of the initial relative configurations of the robot vehicle and the goal frame. To carry out this characterization, there is no loss of generality in assuming that our goal frame is as depicted and has coordinates $x_r = 0$, $y_r = 0$, and $\theta_r = 0$. With this set-up, the shaded region consists of points lying on the trajectories of the robot vehicle starting at either (x_0, y_0) (as depicted) or at $(x_0, -y_0)$ on the upper edge of the wedge, and having initial heading angle ϕ_0 satisfying $\phi_0 \le \phi_{cr}$ (resp. $\phi_0 \ge \phi_{cr}$ on the upper edge), with $\phi = \phi_{cr}$ being the smallest magnitude solution of $-\theta_0 - \phi = Si(\phi)$. Let β be the magnitude of the angle between the negative x-axis of the goal frame and the bottom edge of the wedge in Figure 3. Note that $\beta = \theta_0 + \phi_0$. Then ϕ_{cr} must satisfy $Si(\phi_{cr}) = -\beta$, and this has no solution if $\beta > 1.851937...$ radians ($\sim 106°$). Moreover, this explicit relationship between ϕ_{cr} and β shows that ϕ_{cr} is a monotonically increasing function of β which attains a maximum value of π when $\beta = 1.851937....$ As β approaches this value from below, there is an increasingly narrow range of initial headings ϕ_0 which lead to trajectories which approach the goal frame with asymptotically finite curvature.

Remark 5. Trajectories in which the vehicle turns more than 2π-radians. Even when $\phi_0 > \phi_{cr}$ in Figure 3 (i.e. when the vehicle frame is too closely aligned with the line-of-sight vector to the goal [=bottom of the wedge]), it is possible to find gains k_1, k_2 such that the resulting trajectory generated by (6) approaches the goal frame with asymptotically finite curvature. This will involve the vehicle turning through more than 2π-radians (See Figure 4.), however, and the motion will be highly sensitive to errors in the measurements of the heading angle ϕ. By restricting the application of the control law (6) to vehicle motions whose initial conditions are confined to a wedge shaped region whose upper and lower edges make respective angles $\pm 1.418151576...$

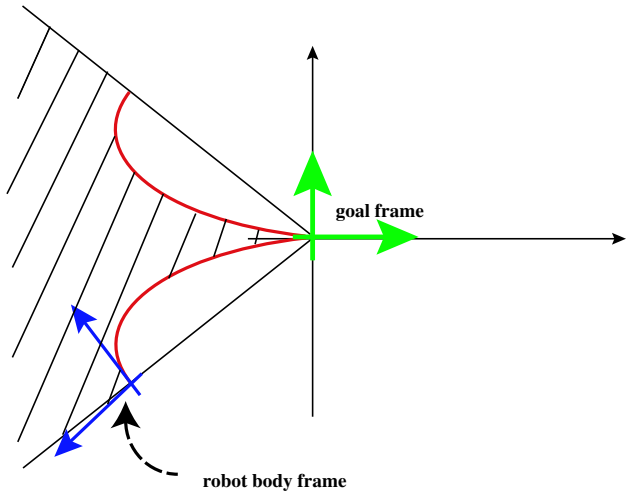

Fig. 3. The "red" trajectories depicted in the figure are "critical" in terms of the path having asymptotically finite curvature; as discussed in the text, we have $k_1/k_2 = 1$. The top red trajectory corresponds to initial heading angle $\phi > 0$ (with corresponding controlled motion turning monotonically counterclockwise), while the lower trajectory has the initial heading $\phi < 0$ as depicted (and the controlled motion has the vehicle turning monotonically in the clockwise direction).

with the negative goal-frame x-axis, it is possible to avoid both undue sensitivity of the control law to the initial heading ϕ_0 as well as situations in which there are multiple solutions to the critical heading equations $Si(\phi_{cr}) = -\beta$. By restricting to a slightly narrower wedge, say $\beta = 1.370762...$, it is possible to have critical heading values less than $\pi/2$ in magnitude. In laboratory settings where most sensors are aimed in the forward direction and have limited peripheral acuity, this is desirable.

3.2 Cases in which the goal frame is not stationary.

As in the previous subsection, let (x_r, y_r, θ_r) be coordinates of the goal frame. Now, however, assume these are varying in time in accordance with the motion control law

$$\begin{pmatrix} \dot{x}_r \\ \dot{y}_r \\ \dot{\theta} \end{pmatrix} = \begin{pmatrix} v_r \cos\theta_r \\ v_r \sin\theta_r \\ \omega_r \end{pmatrix}, \tag{11}$$

(The goal frame here might be thought of as the body frame of the lead vehicle in a leader-follower pair.) Now the relative bearing and distance equations (5) are replaced by

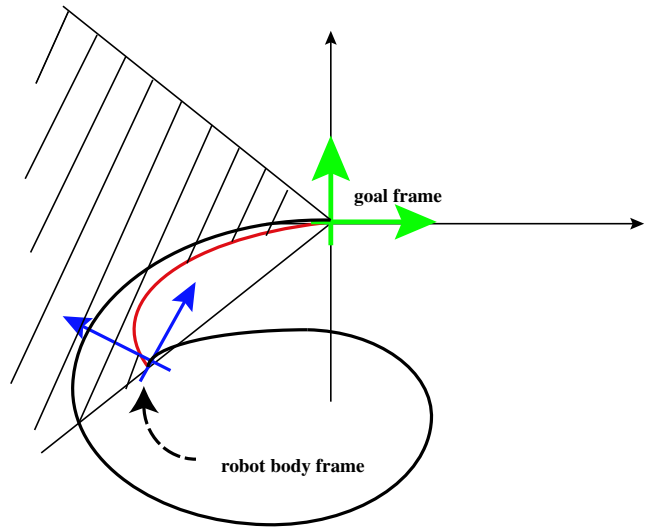

Fig. 4. The "red" trajectory is again "critical" in terms of the path having asymptotically finite curvature; as discussed in the text, we have $k_1/k_2 = 1$. For initial headings aimed at the goal frame with an initial approach angle less than ϕ_{cr}, it is possible in cases where β is not too large to choose gains k_1, k_2 such that the asymptotic curvature of the approach is zero—at the cost of the trajectory turning through more than 2π radians, as depicted here.

$$\begin{pmatrix} \dot{\rho} \\ \dot{\phi} \\ \dot{\alpha} \end{pmatrix} = \begin{pmatrix} -v\,\cos\phi + v_r\cos(\phi-\alpha) \\ -\omega + (v/\rho)\sin\phi - (v_r/\rho)\sin(\phi-\alpha) \\ \omega_r - \omega \end{pmatrix}, \tag{12}$$

where $\alpha = \theta_r - \theta$ is the relative angle between the two frames. For the control law (6)

$$\begin{aligned} v &= k_1\rho, \quad \text{and} \\ \omega &= k_2\phi + k_1\sin\phi, \end{aligned}$$

the closed-loop equations are

$$\begin{pmatrix} \dot{\rho} \\ \dot{\phi} \\ \dot{\alpha} \end{pmatrix} = \begin{pmatrix} -k_1\rho\cos\phi + v_r\cos(\phi-\alpha) \\ -k_2\phi - (v_r/\rho)\sin(\phi-\alpha) \\ \omega_r - k_2\phi - k_1\sin\phi \end{pmatrix}. \tag{13}$$

If we assume ω_r, v_r are both constant, we can find equilibrium solutions to the equations (13). If $k_1, k_2 > 0$ and $\omega_r > 0$ is sufficiently small, there is a unique equilibrium solution to (13):

(i) a unique ϕ_e satisfying $|\phi_e| < \pi/2$ and $\omega_r - k_2\phi_e - k_1\sin\phi_e = 0$, and in terms of this:

(ii) $\rho_e = v_r/\sqrt{k_1^2 - \omega_r^2 + 2k_2\omega_r\phi_e}$, and

$$\alpha_e = \arctan_2[k_1^2 - \omega_r(\omega_r - k_2\phi_e), \omega_r\sqrt{k_1^2 - (\omega_r - k_2\phi_e)^2}].$$

If $0 < k_2 < k_1$, then it is not difficult to show that for certain ranges of the parameter ω_r, there will be multiple equilibrium solutions to (13). To see this, we rewrite the solutions (ii) as

$$\begin{aligned} \rho_e &= v_r/\sqrt{k_2^2\phi_e^2 + k_1^2\cos^2\phi_e}, \quad \text{and} \\ \alpha_e &= \arctan_2[-k_2\phi_e\sin\phi_e + k_1\cos^2\phi_e, \omega_r\cos\phi_e]. \end{aligned}$$

Consider the equilibrium equation

$$\omega_r - k_2\phi_e - k_1\sin\phi_e = 0.$$

Setting the derivative of the left hand side equal to zero yields

$$k_2 + k_1\cos\phi_e = 0,$$

and normalizing $k_1 = 1$ and eliminating the variable ϕ_e from the equations yields the bifurcation locus in terms of ω_r and k_2. This is depicted in

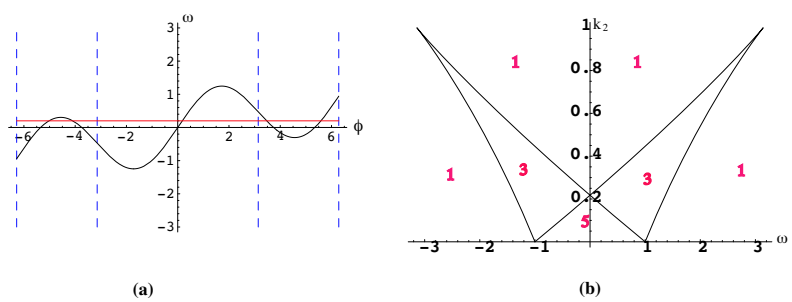

Fig. 5. Bifurcations of leader-follower solutions. In (ϕ_e, ω_r)-space, Figure (a) shows how solution multiplicities for relative equilibria for the control law (6) change as we vary k_2 and ω_r. In terms of the parameters ω_r and k_2, Figure (b) shows the corresponding numbers of relative equilibrium solutions in each part of the parameter domain $-\pi \le \omega_r \le \pi$, $0 \le k_2 \le 1$.

Figure 5(b). Thinking in terms of the right-half of the (ω, k_2)-plane as shown, when ω_r is fixed (at say $\omega_r = \pi/2$) and $k_2 \approx 1$, there is one and only one solution ϕ_e to the equation $\omega_r - k_2\phi k_1\sin\phi = 0$. As the gain parameter k_2 is decreased in value, the unique equilibrium solution ϕ_e increases slightly in magnitude, until we come to a critical value (approximately $k_2 = 0.58271$) at which a *saddle-node* bifurcation occurs in the equilibrium solutions to (13),

and for values of k_2 slightly smaller than this critical value, there are three equilibrium solutions, one of which lies in the range $0 < \phi_e < \pi/2$ and two of which take values $\phi_e \approx 3\pi/2$. The solutions resulting from the bifurcation correspond to the robot vehicle (follower robot) tracking the moving frame (leader robot) along circular paths which lie strictly inside the path of the leader. One of the equilibria has the follower slightly behind the leader, and the corresponding linearized motion (13) is stable (i.e. has all left half-plane eigenvalues), while the other bifurcated equilibrium has the follower slightly ahead of the leader, with the corresponding linearized motion being unstable. Both these equilibrium solutions have the follower on a circular trajectory which lies inside the circular trajectory being followed by the leader. The equilibrium motion defined by $0 < \phi_e < \pi/2$ has the follower also slightly behind the leader, tracing a circle of larger radius, however. The linearized equation of motion are stable for this solution.

If both ω_r and k_2 are decreased further, an additional saddle-node bifurcation occurs, resulting in a total of five relative equilibrium solutions to (13). These solutions have the follower tracing circles of radius larger than that of the leader. One solution (unstable) has the follower ahead of the leader, while the other solution is stable with the follower behind the leader.

When $k_2 > k_1$, there is a unique (one-to-one) correspondence between leader-frame turning rates ω_r and equilibrium bearings ϕ_e. For any given k_2, however, some values of ω_r will cause the linearized motion of (13) about the relative equilibrium to be unstable. (The leader will be turning too sharply for the follower to keep up!) Nevertheless, one can show that for any ω_r, there will be choices of k_2 such that $-\pi/2 < \phi_e < \pi/2$, and such that the motion (13) is stable.

In light of the above considerations and calculations, the following proposition summarizes the dependence of the control law (6) on the relative magnitudes of the forward motion gain k_1 and the turning gain k_2.

Proposition 1. *Let (6) be normalized by setting $k_1 = 1$. Let v_r and ω_r be respectively the constant forward velocity and constant turning velocity of the "goal frame" (=leader robot). If $k_2 > 1$, and $-1 - \pi/2 \leq \omega_r \leq 1 + \pi/2$, then there is a unique relative equilibrium*

$$\begin{aligned} &\phi_e \text{ s.t. } -\pi/2 \leq \phi_e \leq \pi/2, \quad \text{with} \quad \omega_r = k_2\phi_e + k_1 \sin\phi_e \\ \rho_e &= v_r/\sqrt{k_2^2\phi_e^2 + k_1^2\cos^2\phi_e}, \quad \text{and} \\ \alpha_e &= \arctan_2[-k_2\phi_e \sin\phi_e + k_1\cos^2\phi_e, \omega_r \cos\phi_e]. \end{aligned}$$

Moreover, for any leader turning velocity ω_r, there is a value of follower turning gain $k_2 > 1$ such that a unique stable relative equilibrium motion of this form exists with $|\phi_e| < \pi/2$.

Remark 6. Having studied the close-loop feedback (6) for both the cases in which to goal frame is moving and stationary, it is interesting to note that precisely the same condition, $k_2 > 1$, which guarantees trajectories with asymptotically finite curvature in the case of a stationary goal frame also guarantees unique stable relative equilibrium motions in the case in which the goal frame is undergoing a constant velocity circular motion.

3.3 Simple vehicle motion to a preassigned distance to a goal ending with orientation perpendicular to the line of sight vector:

We shall next define a feedback control law of the same form which steers the vehicle from all but a thin set of initial configurations asymptotically toward a final configuration in which it is a prescribed distance d from a goal point and aligned in a direction perpendicular to the line-of-sight vector fro the origin of the robot's body frame to the goal point. Let ρ and ϕ be the sensor-measured quantities depicted in Figure 1(b). Let d be a desired distance from the goal point (x_r, y_r) to which we wish to have the robot vehicle approach. We shall assume without loss of generality that $(x_r, y_r) = (0, 0)$. Consider the following control law:

$$\begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{pmatrix} = \begin{pmatrix} \lambda(\rho - d)\cos\theta \\ \lambda(\rho - d)\sin\theta \\ \rho\sin\phi - d \end{pmatrix}, \tag{14}$$

In applying this control laws, we assume the initial distance $\rho_0 > d$. We then have the following.

Lemma 1. *Along trajectories of (15),*

$$\begin{aligned} &\dot{\phi} = \lambda(1 - \tfrac{d}{\rho})\sin\phi - (\rho\sin\phi - d), \ \textit{and} \\ &\dot{\rho} = -\lambda(\rho - d)\cos\phi. \end{aligned} \tag{15}$$

The next proposition provides important information characterizing the motions of (14) (or equivalently (15)).

Proposition 2. *The variety*

$$\rho\sin\phi - d = 0$$

is an invariant set for the system (14).

Proof: Let x_0,y_0, and θ_0 be intial conditions for (15) such that with

$$\rho_0 = \sqrt{(x_0 - x_r)^2 + (y_0 - y_r)^2}, \quad \text{and}$$

$$\phi_0 = \arctan(x_r - x_0, y_r - y_0),$$

we have $\rho_0 \sin\phi_0 - d = 0$. Let x,y, and θ evolve according to (14), and let

$$\rho(t) = \sqrt{(x(t) - x_r)^2 + (y(t) - y_r)^2}, \quad \text{and}$$

$$\phi(t) = \arctan(x(t) - x_0, y(t) - y_0).$$

Introduce the new variable

$$s(t) = \rho(t)\sin\phi(t) - d.$$

The time derivative with respect to t is

$$\dot{s}(t) = \dot{\rho}\sin\phi + \rho\cos\phi\dot{\phi},$$

and from the preceding lemma, it follows that this may be rewritten

$$\dot{s}(t) = -\rho(t)\cos\phi(t)s(t).$$

We thus have the following representation of s:

$$s(t) = e^{-\int^t \rho(\sigma)\cos\phi(\sigma)\,d\sigma} s_0. \tag{16}$$

This proves the proposition, since $s_0 = 0$ implies $s(t) \equiv 0$. □

The geometry of the invariant variety may be described in detail as follows. First, note that for each $\rho > d$, there are two solutions ϕ_1, ϕ_2 to the equation $\rho\sin\phi - d = 0$. These are illustrated in Figure 6. The red vectors

are aligned with the vehicle body-frame x-axis, corresponding to the direction of forward motion of the vehicle. one is pointed toward the circle of radius d about the goal point along a line which is tangent to the circle. The other is pointed away from the same circle along a line which is also tangent. From the point (x, y), if the vehicle undergoes motion along the line in the direction of the vector pointed toward the circle, maintaining its direction in alignment with the line, it is clear that $\rho(t)\sin\phi(t) - d \equiv 0$. Similarly, along the line in the direction of the vector pointed away from the circle, we also have $\rho(t)\sin\phi(t) - d \equiv 0$.

These observations provide a complete picture of the invariant variety, $\rho\sin\phi - d = 0$. For any location (x, y) of the vehicle frame in the region $\rho > d$, the point set $\rho\sin\phi - d = 0$ consists of two components: (i) a line segment from (x, y) which is tangent to the circle $\rho = d$ (with corresponding vehicle motions (14) directed along the line toward the circle), and (ii) a semi-infinite line from (x, y) directed away from the circle, making an angle with the radial line segment from the goal frame origin to (x, y) which is supplementary to the angle made by the first line segment. The vehicle motions under (14) along this line segment are directed away from the circle. To distinguish these two components, we call the first the *stable branch* and the second the *unstable*

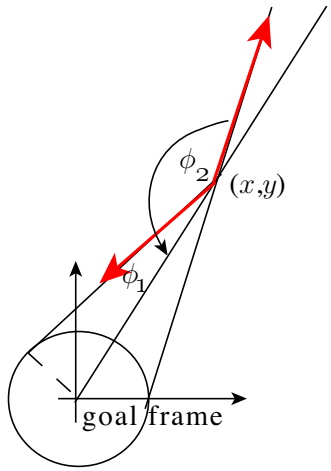

Fig. 6. The variety $\rho \sin\phi - d = 0$ has two components. The point labeled (x, y) is ρ units from the origin of the "goal frame." The red arrows represent the directions of the x-axis of the robot frame with line of sight directions ϕ_1 and ϕ_2 toward the origin of the goal frame. Both these direction segments, if extended, tangentially intersect the circle of radius d centered at the origin of the goal frame.

branch of $\rho \sin\phi - d = 0$. This invariant set is represented by the blue lines in Figure 6.

The v-shaped locus of points $\rho \sin\phi - d = 0$ separates the region of the plane outside the circle $\rho = d$ into two regions: $\rho \sin\phi - d > 0$, points of which lie inside the "v", and $\rho \sin\phi - d > 0$, containing the points outside (on the convex side of) the "v". If the vehicle is started at a point (x, y) with any heading satisfying $\rho \sin\phi - d > 0$ (direction vector inside the "v"), then, from (16), $\rho \sin\phi - d > 0$ for all t, and from (14) we see that the vehicle follows a path which curves counterclockwise, toward the goal circle. To prove that it actually approaches being tangent to the circle, we shall use the following two lemmas.

Lemma 2. *Suppose the forward velocity gain λ is in the interval $0 < \lambda < d$. Then if the initial value of ρ is given by*

$$\rho_0 = \sqrt{(x_0 - x_r)^2 + (y_0 - y_r)^2},$$

the set E defined by the inequalities $-\pi/2 \le \phi \le \pi/2$, $d \le \rho \le \rho_0$ is invariant under the motion of (15).

Proof: From Lemma 1,

$$\dot{\phi} = \lambda(1 - \tfrac{d}{\rho}) \sin\phi - (\rho \sin\phi - d), \text{ and}$$
$$\dot{\rho} = -\lambda(\rho - d) \cos\phi.$$

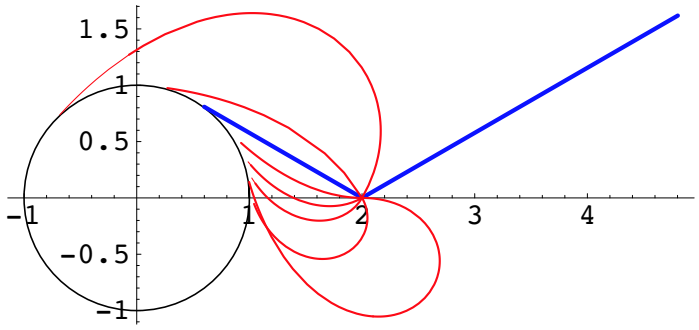

Fig. 7. Typical vehicle trajectories associated with motion control law (14), all starting at the point whose goal-frame coordinates are $(x, y) = (2, 0)$. There are three invariant sets associated with this law: $\rho \sin\phi - d > 0$, $\rho \sin\phi - d = 0$, and $\rho \sin\phi - d < 0$. The two branches of $\rho \sin\phi - d = 0$ are depicted in blue.

We shall prove that almost all trajectories of (15) which start on the boundary of E are directed into the interior of E. All other trajectories whose initial conditions place them on the boundary of E remain on the boundary for all time.

From (15), it follows that $\rho(t)$ is decreasing throughout $d < \rho \leq \rho_0$, provided $-\pi/2 < \phi < \pi/2$. If $\phi = \pi/2$, we find from (15) that $\dot{\phi} < 0$, provided $\rho > d$. If $\phi = -\pi/2$, on the other hand, $\dot{\phi} > 0$. When $\rho = d$, $\dot{\phi} \geq 0$, and ϕ approaches $\pi/2$. This argument has shown that all trajectories starting at points on the boundary of E <u>other than</u> the set $\rho = d$ are directed into the interior of the set. The set $\rho = d$, $-\pi/2 \leq \phi \leq \pi/2$ is itself invariant, as is easily shown. $\square$

Consider the function

$$V(\rho, \phi) = \frac{1}{2} (\rho \sin\phi - d)^2$$

on the ivariant set E defined in the previous proposition. Differentiating this along trajectories of (15), we obtain (using Proposition 2)

$$\dot{V} = -\rho \cos\phi \, (\rho \sin\phi - d)^2.$$

Since the compact set E is invariant, and since solutions of (15) tend in the limit $t \to \infty$ to approach the zero set of $\dot{V}$ (Lasalle's theorem), we conclude that in the asymptotic limit, trajectories must approach either $\cos\phi = 0$ of $\rho \sin\phi - d = 0$. We know from Proposition 1 that $\rho \sin\phi - d = 0$ is invariant under the motion of (15). Moreover, there is no solution (15) such that $\rho \cos\phi \equiv 0$ while $\rho \sin\phi - d \neq 0$.

From the ρ-component of equation (15) we know that in the set E, $\rho(t)$ decreases in the limit to d. (If $\rho \to \rho_* > d$, $\phi(t)$ could not be confined ti a

small neighborhood of $\pi/2$, so that $\rho(t)$ could not remain bounded away from d.) Then since $\rho(t) \to d$, the corresponding limit for $\phi(t)$ must be $\pi/2$, and we have established the following.

Lemma 3. *Suppose the forward velocity gain λ is in the interval $0 < \lambda < d$, and let ρ_0 be defined as in Proposition 2. Any motion of (15) which starts in the set $E = \{(\rho, \phi) : d \leq \rho \leq \rho_0, -\pi/2 \leq \phi \leq \pi/2\}$ tends asymptotically to $(\rho, \phi) = (d, \pi/2)$. I.e., the vehicle coordinates (x, y, θ) are such that (x, y) tends to a point on the circle*

$$(x - x_0)^2 + (y - y_0)^2 = d^2$$

with the vehicle's body-frame x-axis, $(\cos\theta, \sin\theta)$ tangent to the circle, pointed in the counter-clockwise direction along the circle.

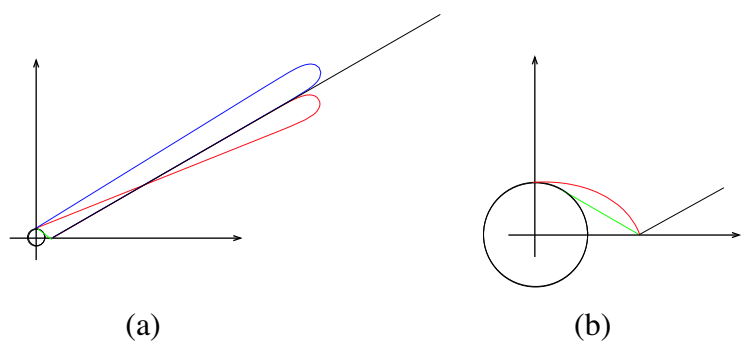

Fig. 8. The feedback control (14) is again *almost* asymptotically stabilizing. Depicted here are several trajectories with initial coordinates $x = 0.5$, $y = 0$, and various values of initial orientation θ. The goal is to move toward a circle of radius 1 centered at $(x, y) = (-1.5, 0)$ with final alignment tangent to the circle, and pointed in the counterclockwise direction of tangency. Figure (a) shows two trajectories in which the initial conditions have the vehicle slightly misaligned in the clockwise (red) and counterclockwise (blue) sense with respect to the unstable branch of the invariant variety $\rho \sin\phi - d = 0$. Figure (b) shows two trajectories near the stable branch of the invariant curve $\rho \sin\phi - d = 0$. Indeed the green trajectory is coincident with this stable branch.

The following analysis shows that in fact the proposed control law

$$v = \lambda(\rho - d), \qquad \omega = \rho \sin\phi - d$$

accomplishes the desired objective for (14) for all initial conditions except those on a very thin set.

Proposition 3. *Consider the controlled system (14) with initial conditions* (x_0, y_0, θ_0)*. Suppose* λ *is in the interval* $0 < \lambda < d$*, and let*

$$\rho_0 = \sqrt{(x_0 - x_r)^2 + y_0 - y_r)^2}, \quad \text{and}$$

$$\phi_0 = \arctan 2[x_r - x_0, y_r - y_0].$$

If the initial configuration of the vehicle body frame (x_0, y_0, θ_0) *is not directed along the unstable branch of* $\rho \sin\phi - d = 0$*, then the motion of (15) tends asymptotically to* $(\rho, \phi) = (d, \pi/2)$*. The corresponding* (x, y, θ) *trajectory generated by (14) tends asymptotically to a point on the circle*

$$(x - x_r)^2 + (y - y_r)^2 = d^2$$

with the body-frame x*-axis [in the direction* $(\cos\theta, \sin\theta)$*] tangent to the circle, pointed in the counterclockwise direction.*

Proof: There are three cases to consider: $\rho_0 \sin\phi_0 - d > 0$, $\rho_0 \sin\phi_0 - d < 0$, and $\rho_0 \sin\phi_0 - d = 0$. In the case $\rho_0 \sin\phi_0 - d > 0$, let ϕ_l denote the larger of the two solutions of $\phi = \arcsin[d/\rho_0]$ in $(\frac{\pi}{2}, \pi)$. While $\pi/2 < \phi < \phi_l$, it follows from (16) that $\rho \sin\phi - d$ is increasing. The ρ-component of equation (15) shows that ρ is also increasing. Using the ϕ-component of equation (15), it can then be shown that (after a possible initial increase) ϕ decreases and passes through the value $\pi/2$. As noted above, when $\phi = \pi/2$ the right hand side of the ϕ-component of equation (15) is negative. Let t_* be the time at which the trajectory which has initial conditions (ρ_0, ϕ_0) satisfies $\phi(t_*) = \pi/2$. Let $E_* = \{(\rho, \phi) : d \leq \rho \leq \rho(t_*), -\pi/2 \leq \phi \leq \pi/2\}$. By Lemma 2, the trajectory of (15), remains confined to E_* for all $t \geq t_*$, and by Lemma 3, it tends asymptotically to $(\rho, \phi) = (d, \pi/2)$, proving the proposition in this case.

In the case $\rho_0 \sin\phi_0 - d < 0$, we shall again show that the trajectory enters an invariant set of the form characterized in Lemma 2. If $\phi_l < \phi_0 < \frac{3\pi}{2}$, then a careful examination of the right-hand side of the ϕ-component of equation (15) shows that $\dot{\phi}$ is strictly positive. Hence, $\phi(t)$ increases through the value $3\pi/2$ ($\sim -\pi/2$). Let t_* be the time such that $\phi(t_*) = -\pi/2$. As above, the trajectory remains in the invariant set $E_* = \{(\rho, \phi) : d \leq \rho \leq \rho(t_*), -\pi/2 \leq \phi \leq \pi/2\}$ for all $t > t_*$, and by Lemma 3 tends asymptotically toward the point $(\rho, \phi) = (d, \pi/2)$, proving the proposition in the case that $\rho_0 \sin\phi_0 - e < 0$.

When $\rho_0 \sin\phi_0 - d = 0$, there are two subcases to consider: $\phi) \leq \pi/2$ and $\phi_0 > \pi/2$. In the latter case, we find from equation (8) that $\rho(t)$ grows without bound. The motion is along the unstable branch of $\rho \sin\phi - d \equiv 0$, and this case is ruled out under our hypothesis. On the other hand, when $\phi \leq \pi/2$, the motion is along the stable branch of $\rho \sin\phi - d \equiv 0$. From equation (8) we find that $\rho(t)$ is decreasing and must approach the point $(\rho, \phi) = (d, \pi/2)$ as claimed. This proves the theorem. □

A completely symmetric analysis may be applied to the feedback law $v = \lambda(\rho - d)$, $\omega = \rho \sin\phi + d$, which produces trajectories tangentially ap-

proaching the circle of radius d about the goal point in the clockwise direction of tangency.

4 Compositions of motions using simple motion primitives for nonholonomic vehicles

A small catalogue of control laws of the type analyzed in detail in the preceding section appears in Table 1. The common feature of these laws is that they are based continuous feedback with singularities, and as such they achieve their objectives only in certain operating domains, while failing to work as desired on "thin" sets in the state space. While important qualitative features of motion segments generated in this way are becoming increasingly well understood, global strategies for concatenating such segments to achieve complex motion objectives remain under development. Path planning for both single robots and formations can be carried out using simulated motions based on compositions of feedback laws like the ones listed in the table. Such laws with appropriately chosen parameters will be assembled in a sequence in which care is taken to ensure that each motion segment terminates inside the domain of stable operation of the feedback law chosen as a successor in this sequence. Global figures of merit such as speed of transit, minimum excursion from a desired path, minimum expenditure of actuator energy, and so forth can be employed to find paths that meet a variety of operational objectives. Preliminary work on such planning will be reported elsewhere.

Table 1

A Small Catalogue of Feedback Control Laws of the Form (5) Using Real-Time Relative Position Data from Sensors

	Control Law	Objective	Analysis
1.	$v = k_1\rho$ $\omega = k_2\phi + k_1 \sin\phi$	Choose gains k_1 and k_2 to move vehicle into alignment with a prescribed frame	Sections III A. (stationary frame) and III B (goal frame in motion).
2.	$v = k_1\rho$ $\omega = k_2(\phi - \phi_0)$	Choose gains k_1 and k_2 to move vehicle to follow a moving frame with prescribed offset bearing ϕ_0.	See Baillieul & Suri, 2003
3.	$v = k_1(\rho - d)$ $\omega = k_2(\rho \sin\phi - d)$	Choose gains k_1, k_2 to move vehicle to a preassigned distance d from a goal ending with orientation perpendicular to the line of sight vector.	Section III C.
4.	$v = k_1(\rho - d)$ $\omega = k_2(\rho \sin\phi + d)$	Same as previous law, but steering vehicle to orientation rotated 180°.	C.f. Section III C.
5.	$v = 1$ $\omega = 1/d + k_1(\phi - \pi/2)$ $+k_2(\rho - d)$	Choose gains k_1, k_2 to steer vehicle onto a path coincident with a circle of radius d about a goal position.	No published analysis.

Along with such theoretical investigations, a number of implementation-related problems and challenges are also under investigation. While we can find necessary and sufficient conditions for stably rigid motions of formations of point robots in the plane in a number of places in the recent literature ([1],[10],[11], and [18]), the translation of these results to the case of formations of nonholonomically constrained robot motions remains work-in-progress. For the case in which a formation is comprised of two robot vehicles—say a leader

and a follower—the domain of stable operation of control laws such as 1 or 2 in the table below is fairly straightforward to determine. For the case in which such laws are spatially cascaded for controlling large scale formations, however, care must be exercised that no vehicle in the formation is forced outside its range of stable operation. Another problem, which is beyond the scope of the present discussion is that accurate relative position and bearing measurements may be difficult to achieve in actual implementations. In our own laboratory experiments, special markings have been used on each robot in the formation so that it can be unambiguously identified by the sensors on its formation team mates. In the field, such signature markings will not typically be available—especially in the case of operations in adversarial environments. A major open area for future research is the development of cooperative control strategies involving tracking with stochastic, unreliable, and even deliberately falsified sensor data.

5 Conclusions

The paper has provided a detailed analysis of the qualitative features of several continuous feedback laws for motion control of nonholonomic vehicles. Such feedback laws are associated with sets of singular states and parameters in which they fail to achieve desired objectives, but we have shown that in many cases such singular sets can be avoided. Guided by the kind of detailed analyses provided in Section III, it is possible to employ sequences of such continuous feedback laws with appropriate strategies for switching from one control to the next so that overall motion objectives are met while singular or nearly singular operating conditions are avoided. How well such switching strategies can be made to work as the number of robots in a formation becomes large remains under investigation. Related work is now under way to understand the effects of noise, sensor errors, feedback intermittency, and other implementation issues involving the growing catalogue of *feedback-with-singularities* control designs of the type discussed in the paper. It is increasingly of interest in the development of applications to consider the use of feedback data acquired from arrays of spatially distributed and wirelessly networked sensors. Current research is aimed at understanding how the performance of the types of control laws introduced here changes when the feedback signals are quantized and subject to delays and intermittent disruptions. Preliminary results on such questions are reported in [2] and [3].

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